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Large language models and AI agents in disaster-resilient infrastructure: concepts, applications, pathways, and challenges

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ABSTRACT

This paper presents a systematic framework for integrating large language models (LLMs) and AI agents into disaster-resilient infrastructure research. By examining seven key task domains—knowledge anchoring, prediction, assessment, intelligent design and simulation, data augmentation, embodied intelligence, and decision support—we show how LLMs and AI agents can reshape traditional disaster-resilience workflows through enhanced multimodal perception, long-horizon reasoning, cross-stage coordination, and automated task execution. Two representative case studies further demonstrate the feasibility of this paradigm: (1) a multimodal building-damage assessment system capable of city-scale semantic reasoning and transparent diagnostic interpretation, and (2) a LangGraph-based reasoning agent for automated seismic resilience assessment and recovery decision-making of infrastructure systems. These cases highlight the potential of LLMs and AI agents to bridge fragmented data sources, orchestrate multi-stage analytical workflows, and support high-impact resilience applications. Despite these advances, key challenges remain, including data scarcity, reliability and consistency issues, deployment constraints, and ethical or security concerns. Addressing these limitations is essential for developing trustworthy, efficient, and scalable intelligent disaster-prevention systems. Overall, this work establishes a unified cognitive–action paradigm that offers forward-looking pathways for the next generation of intelligent and resilient infrastructure technologies.

1. Introduction

As the impacts of global climate change intensify, extreme natural disasters are occurring with increasing frequency and severity, posing unprecedented challenges to the reliability of infrastructure system [1]. Data from the United Nations Office for Disaster Risk Reduction show that cumulative global economic losses from natural disasters between 2020 and 2024 exceeded USD 1.2 trillion, with over 70 % directly attributed to infrastructure damage. In response, researchers have actively explored innovative and efficient pathways for disaster mitigation to reduce the impact of such events and minimize human losses [2]. However, existing technological systems for infrastructure disaster prevention and mitigation remain inadequate, marked by limited forecasting accuracy, delayed emergency response, inefficient decision-making, and a lack of intelligent operational capabilities. There is an urgent need to develop a more efficient and intelligent technological framework to enhance the overall safety and reliability of infrastructure.

In recent years, disaster-resilient infrastructure have garnered growing attention from the global academic community. Disaster resilience refers to the capacity of infrastructure to maintain or rapidly restore functionality when subjected to hazardous events [3]. With the advancement of artificial intelligence (AI), traditional methods grounded in empirical knowledge or physical modeling are increasingly being integrated with intelligent, data-driven approaches, forming hybrid frameworks that leverage the strengths of both paradigms. Against this backdrop, resilient and intelligent disaster prevention have emerged as one of the most cutting-edge and transformative research frontiers in infrastructure safety and reliability.

Propelled by major advances in big data, deep learning (DL), and computational power, next-generation AI technologies—large language models (LLMs) and AI agents—have evolved rapidly, catalyzing a research paradigm shift. LLMs demonstrate strong capabilities for knowledge transfer, often requiring minimal task-specific retraining and performing well even under low-data conditions [4]. In parallel, AI agents—typically incorporating LLMs as their core ‘cognitive

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engines'—integrate autonomous capabilities such as perception, reasoning, planning, memory, and tool use, thereby enabling the automated execution of complex tasks [5]. To date, LLMs and AI agents have been widely applied across diverse domains—including healthcare, financial technology, agricultural science, chemical research, remote sensing, industrial manufacturing, fault diagnosis, and autonomous driving—demonstrating increasingly strong cross-domain adaptability and effective task-solving performance [6–14].

Although LLMs and AI agents have been widely adopted across many domains, their application in the infrastructure disaster resilience field (IDRF) is still at an early stage. A systematic cognitive framework is urgently needed to elucidate the technological evolution of LLMs and AI agents, along with their potential for adaptation and extension. This paper provides a comprehensive research perspective by examining representative application scenarios and corresponding technical implementation pathways (see Fig. 1). We hope that this work offers valuable insights for researchers in IDRF, fosters diverse innovation, and contributes to the advancement and independent development of technologies in this field. The main contributions are summarized as follows:

(1) We establish a unified and paradigm-shifting conceptual framework that systematically integrates large language models (LLMs) and AI agents into disaster-resilient infrastructure research, transforming conventional data-driven and model-centric approaches into an agent-based resilience intelligence paradigm.
(2) We provide a comprehensive and structured synthesis of application scenarios and technical pathways by mapping LLMs and AI agents to seven representative research tasks across the full disaster-resilience lifecycle—from knowledge anchoring and predictive modeling to assessment, intelligent design, embodied intelligence, and decision support.
(3) We critically identify key challenges and limitations hindering the practical adoption of LLM- and agent-based systems in infrastructure disaster resilience, including data scarcity, reliability and uncertainty, deployment cost, ethical and fairness concerns, and operational feasibility, and outline forward-looking research directions toward trustworthy, lightweight, and deployable intelligent resilience systems.

2. Related work

2.1. Status and challenges: resilience and AI -driven infrastructure safety and reliability

The development of infrastructure safety and reliability technologies under disaster conditions can be broadly categorized into four stages: an early phase of passive response, a systematized mitigation stage dominated by engineering defenses, a technology-enabled stage marked by the integration of information technologies, and the current resilience-oriented stage driven by AI (see Fig. 2). 'System robustness, functional integrity, and recoverability' are widely recognized as the core attributes of disaster-resilient infrastructure [15]. These attributes reflect the interdisciplinary nature of resilience-oriented technologies, which rely on knowledge from civil engineering, disaster science, operations research, materials science, and management science. Cross-domain integration supports essential tasks across all disaster phases,

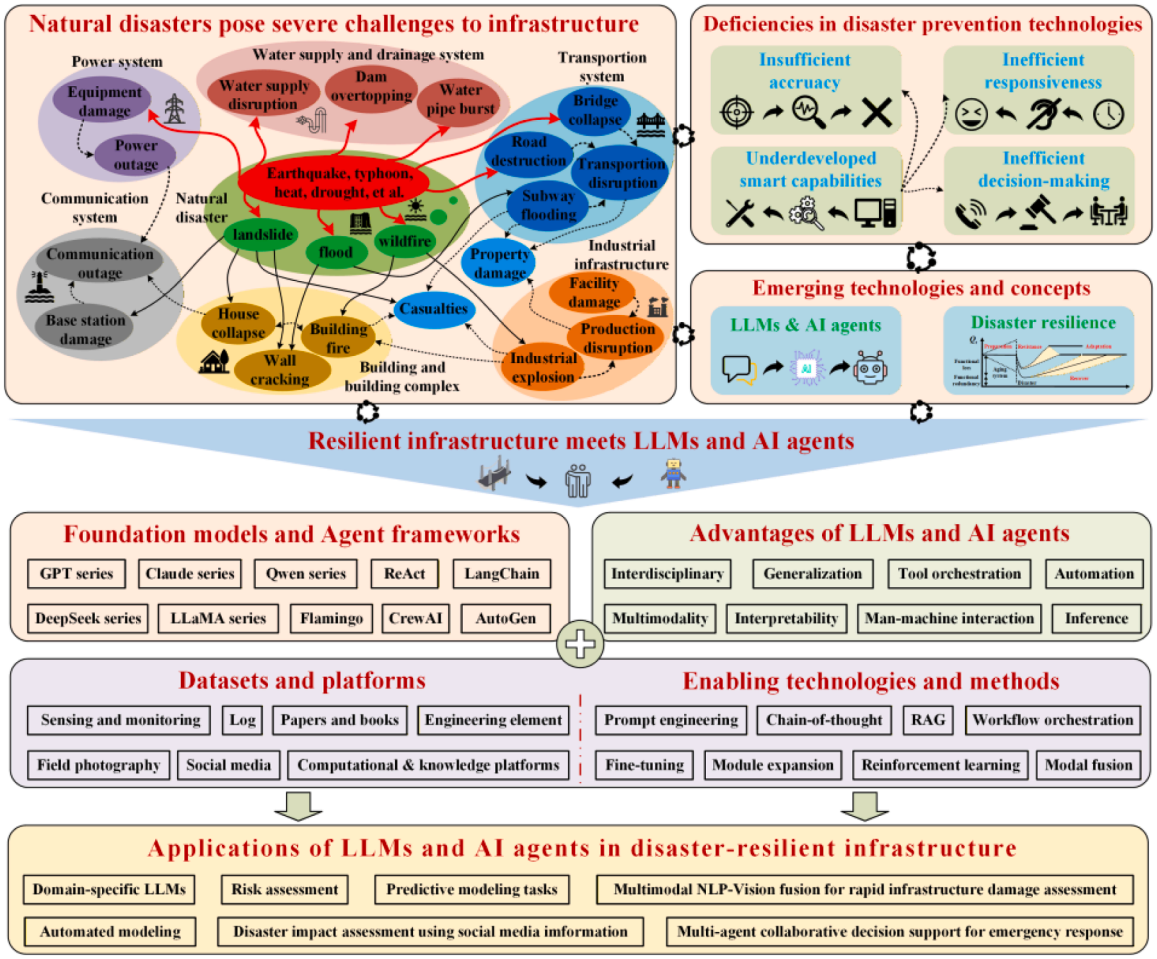

Fig. 1. Overview of the core themes and contributions of this paper.

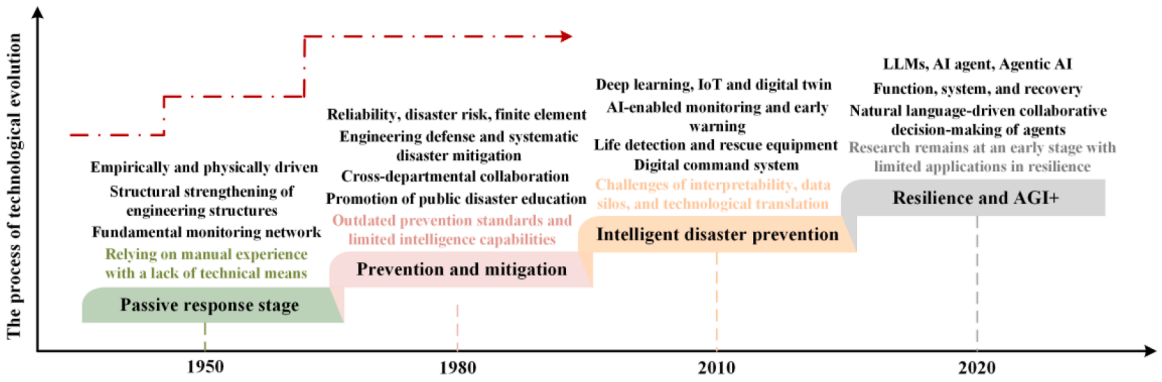

Fig. 2. Development and evolution of infrastructure disaster prevention technologies.

including risk quantification, failure analysis, reliability assessment, emergency response optimization, and resilience design. Since Bruneau et al. introduced the 'resilient city' concept [16], the theoretical framework for resilient infrastructure has continued to evolve [17–19]. However, applying these advances in real engineering practice remains a major challenge and a key direction for future work.

In recent years, the rapid advancement of AI technologies has introduced novel analytical tools and technical approaches to the study of the IDRF. Machine-learning (ML) techniques such as random forests and support vector machines are increasingly used in disaster reliability assessment, where they learn nonlinear structural damage patterns from high-dimensional parameter spaces, thereby overcoming the limitations of empirical fragility curves and simplified physical models [20]. DL enables efficient and scalable prediction of structural failures and functional degradation by automatically extracting features from images and time-series data, effectively addressing challenges that traditional physics-based methods face when dealing with high-dimensional inputs. Graph neural networks embed topological structures, operational states, and failure interactions into unified graph representations, enabling data-driven modeling of cascading failures and cross-infrastructure dependencies [21–24]. Hooman et al. conducted an in-depth analysis of the unique value of digital twin (DT) in the context of disaster management and smart cities, highlighting how their data-driven decision mechanisms provide scientific evidence for decision-makers and facilitate optimal resource allocation [25]. These studies indicate that traditional AI technologies—represented by DL and DT —have been widely applied in IDRF research, demonstrating advantages such as efficiency, intelligence, real-time performance, and precision.

Unlike many other domains, IDRF scenarios are characterized by high uncertainty, elevated risk, dynamic evolution, multisource heterogeneous data, and strong coupling across multiple stages and temporal scales. These attributes indicate that disaster processes are inherently cross-scale, cross-modal, and cross-phase phenomena within complex systems, thereby introducing unprecedented technical bottlenecks for IDRF research. Therefore, although AI has achieved notable advances over empirical and physics-based approaches in prediction, assessment, and decision-making tasks, model-centric and prediction-oriented AI paradigms remain insufficient to address these inherent challenges.

First, DL models require large volumes of high-quality data, yet real disaster events are infrequent and infrastructure damage records are scarce, limiting the ability to learn reliable disaster patterns and increasing the risk of overfitting. Second, disaster monitoring produces heterogeneous data modalities—sensor time series, remote-sensing imagery, and textual social-media logs—with inconsistent formats and substantial temporal misalignment. DL models and DT are not inherently suited for robust multimodal fusion under such weakly correlated or unaligned data conditions. Third, limited context windows restrict long-horizon reasoning; disaster risk prediction, reliability assessment, and decision-support tasks depend on processing long, dynamic, and heterogeneous information streams that exceed the contextual capacity of DL approaches, hindering their ability to maintain global coherence or capture long-term dependencies. Fourth, the strong spatial variability and diverse operating conditions of infrastructure systems pose significant challenges to the generalization of many deep learning–based models, thereby limiting their robustness and transferability in real-world disaster-prevention scenarios. Fifth, DL model s function as black boxes with limited interpretability, making it difficult to validate their reasoning processes and failing to satisfy the transparency, safety, and auditability requirements of engineering applications. Sixth, suboptimal human—machine interaction remains a barrier, as traditional AI systems often rely on specialized interfaces, rigid command formats, and complex operational workflows that hinder effective use in high-stress and rapidly evolving disaster environments. Finally, challenges in system-wide integration and intelligent coordination persist: disaster-resilience workflows span monitoring, forecasting, assessment, and decision-making, yet most AI systems rely on isolated models without unified data standards or execution frameworks, resulting in weak cross-stage synergy and limited whole-process intelligence.

In light of these challenges, integrating emerging LLM and AI-agent technologies with existing ML, DL and DT offers a promising pathway for overcoming these fundamental limitations and advancing intelligent, reliable, and collaborative disaster-resilience solutions.

2.2. *Comprehensive overview of LLMs and AI agents*

2.2.1. *Foundations and key techniques of LLMs*

A LLM is a machine learning model trained through DL on massive text corpora, capable of generating coherent and natural language text awhile accurately understanding the deep semantic structures embedded in textual data [26]. LLMs are widely applied to a broad range of natural language processing (NLP) tasks, including but not limited to text classification, intelligent question answering, and human–computer dialogue systems. Multimodal large language models (MLLMs) extend these capabilities beyond text, enabling unified understanding across diverse modalities such as images, audio, video, and code [27] This cross-modal integration enables MLLMs to perform tasks such as visual question answering, image captioning, and multi-source information fusion. Although most contemporary LLMs are built upon the Transformer architecture, significant differences exist across Transformer-based models in terms of architectural design, scale, training objectives, and application scope. In particular, encoder-only models (e.g., BERT), encoder–decoder models (e.g., T5), and decoder-only models (e.g., GPT-style LLMs) exhibit fundamentally different capabilities and computational characteristics. While decoder-only LLMs excel at open-ended generation, multi-step reasoning, and agent-based decision-making, encoder-only or encoder-decoder models are often more efficient and effective for

task-specific prediction, representation learning, and deployment under resource-constrained settings. Researchers should select appropriate LLMs according to their specific task requirements.

With the rapid advancement of the field, numerous companies and research institutions have developed large-scale foundation model (FM) systems—such as GPT, Qwen, PaLM, DeepSeek, and the LLaMA series—which are pretrained on massive and heterogeneous datasets through self-supervised learning and can be effectively transferred to a wide range of downstream tasks. The open-source nature of FMs, in turn, accelerates the translation of AI technologies into diverse downstream applications [28]. Supervised fine-tuning (SFT) [29], prompt engineering [30], and retrieval-augmented generation (RAG) [31] represent three widely adopted approaches for enhancing the performance of foundation models in domain-specific tasks. Fig. 3 illustrates the positions of various SFT methods within the decoder-only architecture, examples of carefully designed prompts, and the workflow of RAG.

Overall, LLMs represent a paradigm shift from rule-based or purely statistical learning toward knowledge-centered and adaptive intelligence. Their integration of linguistic generalization, multimodal reasoning, and adaptive flexibility provides a scalable technological foundation for developing intelligent and disaster-resilient infrastructure systems under complex and uncertain environments.

2.2.2. Concept and architecture of AI agents

An AI agent is an intelligent system that interacts with its environment, gathers and processes information, makes decisions, and executes actions to achieve specific objectives [32]. In recent years, the rapid advancement of LLMs has greatly accelerated the development and deployment of AI agent applications. Owing to their strong capabilities in logical reasoning, tool use, strategic planning, and instruction following, LLMs effectively serve as the core cognitive engine of modern intelligent agents [33]. An LLM-based agent is typically implemented through an architecture composed of several core modules—perception, planning, memory, and execution—that jointly enable the model to perform complex, multi-step tasks (see Fig. 4). In essence, an AI agent can be regarded as an integration of an LLM's reasoning capability, planning ability, feedback-interaction mechanism, and tool-driven action execution [34].

The emergence and rapid evolution of frameworks such as LangChain, LangGraph, AutoGPT, and MetaGPT have made the development of specialized AI agents increasingly feasible. When multiple such agents collaborate through communication, coordination, and negotiated decision-making to jointly tackle complex cross-domain problems, the resulting collaborative system is termed Agentic AI. Functionally, Agentic AI resembles an interdisciplinary team of experts, where individual agents assume specialized roles and collaborate synergistically to accomplish complex scientific tasks—exemplifying an advanced form of collective intelligence.

Overall, AI agents provide task-level reasoning and execution capabilities, whereas Agentic AI enables multi-agent collaboration and system-level decision-making. Together, they drive the intelligent and automated advancement of the IDRF' research across the entire chain of perception, assessment, reasoning, and decision-making processes.

2.3. New solution: integration of LLMs, AI agents and prior-generation AI

It is still necessary to conduct a critical comparison between these emerging paradigms (LLM/AI agent) and prior-generation AI (ML/DL/DT, et al.) to elucidate their respective advantages, limitations, and practical feasibility. Table 1 provides a systematic comparison across key dimensions—such as knowledge representation, interpretability, computational efficiency, and deployment feasibilit. Each technological paradigm also has its own distinct benefits and inherent constraints. When a single paradigm fails to meet the performance requirements of risk prediction, reliability assessment, or decision-support tasks, Agentic AI provides an integrative framework that effectively combines emerging and traditional methods to achieve complementary strengths.

Accordingly, employing LLMs—or adopting an Agentic AI paradigm that positions LLMs at the core while integrating ML, DL, and DT as complementary components (LLM-Agentic AI)—has become a necessary pathway to address the key challenges outlined in Section 2.1. First, although LLMs require massive data for pretraining, they can reduce the reliance on large labeled disaster datasets for downstream tasks through generalization and few-shot adaptation, while data augmentation and knowledge-driven reasoning (e.g., retrieval and tool use) further help mitigate the scarcity of real disaster samples. Second, combining LLMs with AI agents provides a unified interface for processing multimodal inputs—including text, sensor time series, remote-sensing imagery, and DT simulation outputs—thereby overcoming the limitations of DT technologies in fusing weakly correlated, multi-scale, and heterogeneous data. Third, the planning and reasoning capabilities of AI agents, when integrated with the linguistic reasoning abilities of LLMs, enable more robust long-horizon analytical capabilities, addressing long-standing challenges in modeling long-chain disaster evolution and cross-stage logical dependencies. Fourth, physics-based and model-centric approaches provide high-fidelity physical constraints and

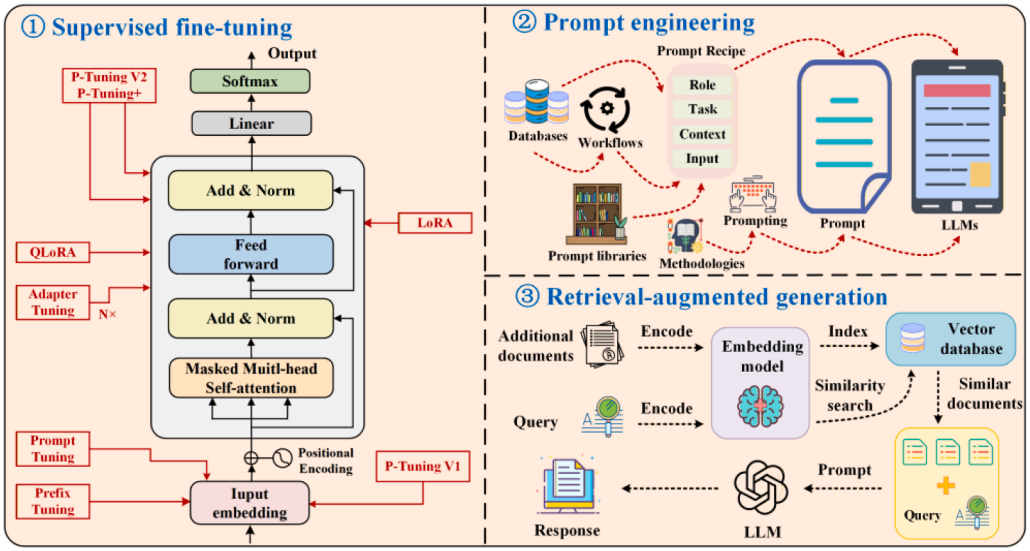

Fig. 3. Common approaches to improving downstream task performance of LLMs.

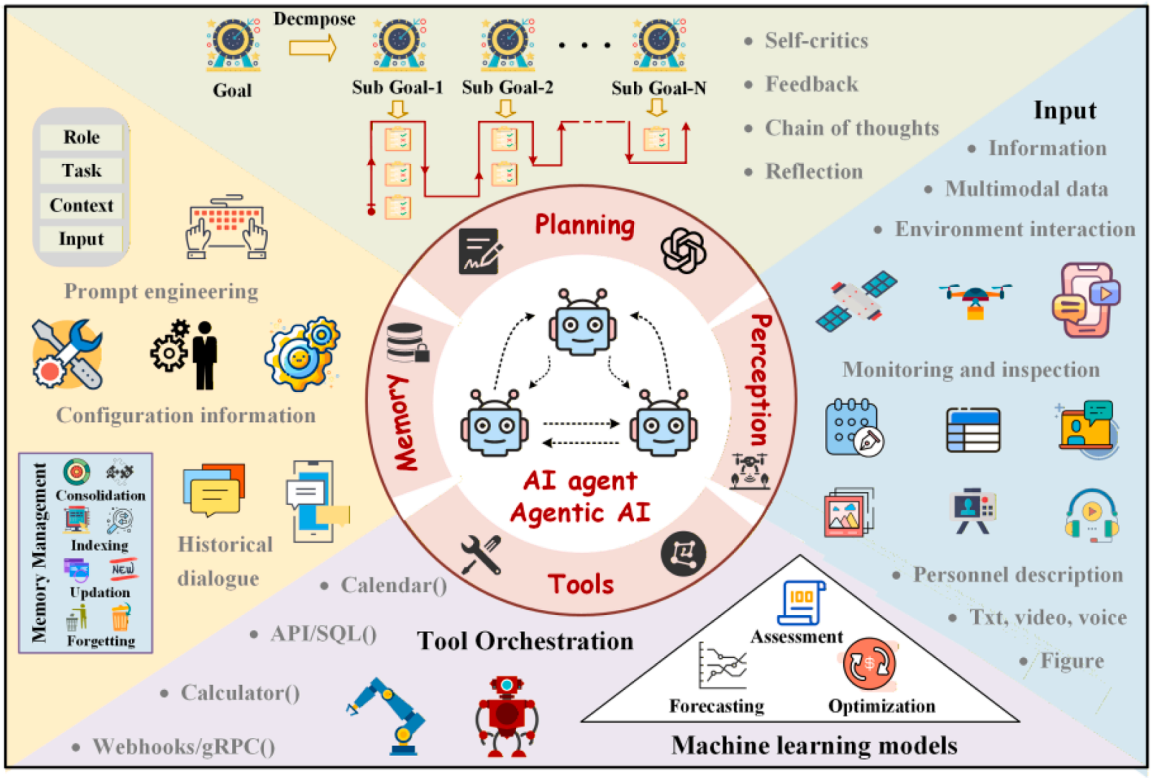

Fig. 4. Fundamental components of LLM-based Agents.

mechanism-based modeling, while LLMs enhance transparency and auditability through interpretable reasoning processes and natural-language explanations, thereby strengthening trust in AI-driven decision-making within IDRF. Fifth, the natural-language interaction capabilities of LLMs greatly improve human–computer interaction, allowing engineers and emergency managers to communicate with and operate the system more intuitively without complex interfaces or rigid command formats. Sixth, the tool-orchestration and workflow-governance capabilities of AI agents enable historically fragmented disaster-prevention processes—such as monitoring, prediction, assessment, and decision-making—to operate collaboratively within a unified architecture, thereby addressing chronic issues of insufficient system integration and weak cross-stage coordination. Finally, LLM-driven embodied intelligence offers a pathway for tightly coupling high-level cognitive decision-making with intelligent physical systems (Internet of Things, unmanned aerial vehicles, and robotics), thereby substantially enhancing the automation, adaptability, and responsiveness required for resilience-oriented disaster prevention.

Overall, LLMs and AI agents offer a systematic pathway that is expected to overcome long-standing bottlenecks in IDRF research. This pathway builds upon the complementary strengths of multiple modeling paradigms: physics-based and numerical models provide high-fidelity physical constraints and quantitative accuracy; deep learning models enable data-driven prediction and pattern extraction; large language models contribute knowledge reasoning and semantic understanding; and AI agents support workflow orchestration, tool invocation, and task execution. By integrating these components into a unified hybrid intelligence framework, it becomes possible to develop end-to-end, explainable, and multimodal disaster-prevention systems with autonomous decision-making capabilities.

3. Potential applications of LLMs and AI agents

3.1. Representative scenarios: targeted seven research tasks in IDRF

We identify seven representative research tasks through which LLMs and AI agents can enhance IDRF applications, with each task aligned with a critical stage of resilience-oriented disaster-prevention workflows. These tasks include domain-knowledge anchoring (Section 3.1.1), predictive modeling (Section 3.1.2), assessment tasks (Section 3.1.3), intelligent design and simulation (Section 3.1.4), data augmentation and multimodal integration (Section 3.1.5), embodied intelligence for disaster-response scenarios (Section 3.1.6), and decision-support tasks (Section 3.1.7). Although these tasks span different functional domains, they collectively form an interconnected pipeline that mirrors the end-to-end IDPF workflow—from sensing and prediction to assessment, planning, and intelligent execution. Together, these seven tasks define the essential research components needed to develop a multi-agent platform for infrastructure resilience management.

3.1.1. General-purpose LLM with IDRF-oriented knowledge anchoring

After domain-knowledge embedding, the foundation model can interpret technical terminology more accurately and substantially reduce domain-specific hallucinations, thereby enhancing safety and reliability in high-risk disaster-prevention scenarios. Moreover, a knowledge-enhanced foundation model not only supports cross-task collaboration and domain-specific reasoning—serving as a prerequisite for constructing AI agent —but also delivers more stable, reliable, and consistent performance in specialized tasks such as structured question answering, professional education, knowledge dissemination, and fairness evaluation. Therefore, developing an open-source, lightweight foundation model equipped with domain expertise is essential for enabling the practical deployment of LLMs and AI agents in the IDRF domain.

Recent efforts to develop domain-specific large models for IDRF

Table 1
Comparison between prior-generation AI and LLM/AI Agent across key technical dimensions.

Aspect	Prior-generation AI	LLMs and AI agents
Knowledge representation	Encodes statistical correlations between inputs and outputs; limited to training data domain.	Learns abstract and contextualized knowledge from massive corpora; strong semantic reasoning; but lacks the structured constraints of domain-specific models.
Data requirements	DL models often require large labeled datasets, some ML methods work well with limited data..	Operate on unstructured data with few-shot or zero-shot adaptation via in-context learning; require massive pretraining data.
Interpretability and transparency	DL models are typically black-box, some ML approaches offer partial interpretability.	Interpretability via reasoning visualization, physics-informed LLMs may balance interpretability and adaptability.
Computational efficiency	DL models require large datasets, some ML methods perform well with smaller datasets.	Very high computational and memory cost during training and inference; requires advanced hardware or cloud resources.
Cross-modal integration	Limited multimodal fusion (e.g., image-text networks).	Natively supports multimodal integration (text, vision, sensor data, code).
Adaptability and generalization	Task-specific; poor extrapolation to unseen domains.	Zero- and few-shot task adaptation, with strong cross-domain transfer and reasoning capabilities.
Reliability and robustness	Sensitive to data quality and overfitting; prone to noise.	May hallucinate or generate inconsistent reasoning; reliability depends on dataset and prompt design.
Deployment feasibility	Widely used in forecasting, recognition, and pattern analysis.	Deployment constrained by hardware, latency, and energy cost.
Human-AI interaction	Interactive dashboards or APIs; limited reasoning dialogue.	Natural language interaction and collaborative decision-making with human-in-the-loop supervision.
Multi-task capability	Specialized models that excel at single tasks; lack cross-task adaptability.	Agentic AI enables integrated workflows across tasks; Reasoning and decomposition of complex tasks.

reflect a growing consensus within the research community to overcome the limitations of general-purpose LLMs and build intelligent systems tailored to real engineering needs. For example, Tongji University released CivilGPT-V2.0, based on the DeepSeek-R1 architecture, which integrates large-scale engineering corpora and domain terminology to significantly improve technical comprehension and engineering-reasoning accuracy [35]. However, its closed-source nature imposes clear constraints on reproducibility, customization, and local deployment. The EngiLLM model developed by Hao's group at Guangzhou University utilizes knowledge-graph-assisted training to structurally integrate academic literature, industry standards, and engineering case studies, demonstrating another effective paradigm for domain-knowledge embedding [36]. Nonetheless, because the model has not yet been publicly released, its training methodology and potential for secondary development remain unverifiable. In addition, the Institute of Geophysics at the China Earthquake Administration fine-tuned the Qwen model using Chinese seismic data and tens of thousands of earthquake-related Q&A pairs to develop 'Earthquake-GPT' [37]. While it shows promise in structured Q&A, education, and science communication, its potential as a general engineering foundation model has not yet been fully demonstrated.

Overall, developing an open-source, lightweight foundation model specific to the IDRF domain has become an urgent priority. However, the unique complexity of IDRF presents several major challenges. First, the field lacks large-scale, high-quality instruction datasets—particularly cross-hazard, multimodal, and multi-stage annotated corpora—making effective supervised fine-tuning difficult. For lightweight models (<10–14B parameters), developing a reliable IDRF-oriented model generally requires at least 50,000–100,000 high-quality, well-structured instruction samples. Manual annotation is a highly challenging task. Second, engineering mechanisms such as disaster dynamics, structural mechanics, and vulnerability theory are difficult to represent in structured forms; existing corpora are fragmented and format-inconsistent, while multimodal data still require expert annotation. Moreover, significant data-sharing barriers exist across agencies, limiting access to multi-organization, multi-scenario datasets. Meanwhile, the absence of internationally recognized benchmark frameworks for IDRF-oriented foundation models makes systematic and comparable performance evaluation difficult.

We propose a reference technical pipeline for developing an initial open-source and lightweight FM tailored to the IDRF domain (see Fig. 5). The foundational corpus consists of domain-specific standards and regulations (e.g., IOS, GB/CECS), academic literature (Elsevier, wiley, springer, et al.), relevant news reports, policy documents (different countries), and professional textbooks. Leveraging the natural-language understanding capabilities of high-performance models such as GPT-5, an Agentic AI system—equipped with data-cleaning, classification, generation, and evaluation modules enabled by prompt engineering—is constructed to process this corpus. Through multi-agent collaboration, the raw corpus is transformed into an SFT dataset enriched with instructions, reasoning chains, and expert knowledge. A sub-14B open-source foundation model is then selected for supervised fine-tuning, yielding a lightweight LLM specifically designed for IDRF applications. Finally, a manually annotated IDRF benchmark dataset is developed to support comprehensive performance evaluation of the specialized model.

3.1.2. LLMs empowering 'predictive' tasks

Predictive tasks—such as disaster-evolution forecasting, risk and failure prediction, and system-level functional-degradation prediction—constitute core components of the IDRF research framework [38–41]. These tasks typically require processing long-horizon information flows that are heterogeneous across multiple scales, modalities, and stages. The emergence of LLMs and AI agents offers a new paradigm for overcoming these bottlenecks and improving the performance of predictive tasks in IDRF.

Enabling LLMs to perform accurate predictions has recently become a major research focus in the academic community and has already achieved promising results across multiple domains [42–45]. Jin et al. demonstrated that LLMs exhibit strong forecasting capabilities for time series data with clear trends and periodic patterns [46]. Sun et al. identified two principal technical pathways: LLM-for-TS and TS-for-LLM, thereby laying the theoretical groundwork for integrative research in this domain [47]. Peng et al. reframed the lane-change prediction task as a language modeling problem by converting heterogeneous driving scenario data into natural language prompts for LLMs. They enhanced the transparency and reliability of predictions by fine-tuning the chain-of-thought reasoning mechanism. They further introduced explainability constraints during the prompting phase, enabling the model to generate a traceable reasoning process alongside the predicted results [48]. This approach represents a pioneering effort in cross-modal knowledge integration and explainable predictive reasoning, offering valuable technical references for the IDRF's research. Nie T et al. formulated the joint estimation and prediction of urban delivery demand as a spatiotemporal learning task driven by graph neural networks, while leveraging LLMs to extract general geospatial knowledge representations from unstructured geographic data [49]. This approach integrates LLMs with DL in a unified manner, effectively overcoming the inherent limitations of conventional DL in

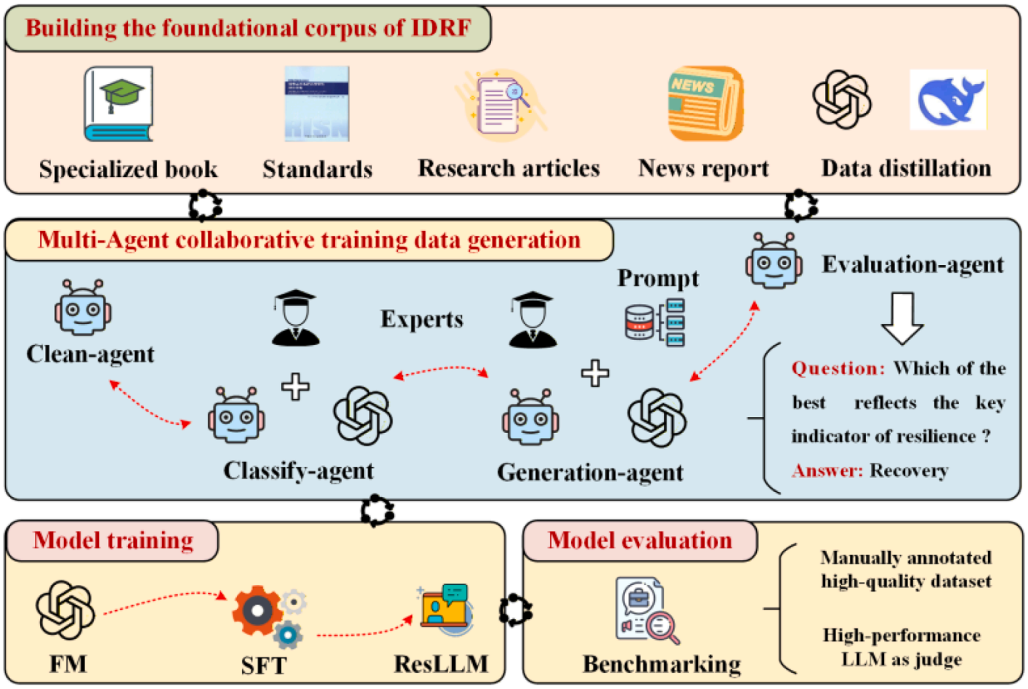

Fig. 5. Technical implementation pathway for the domain-specific LLM.

multisource data integration and cross-modal understanding. Li et al. proposed reframing complex geological prediction tasks as language modeling problems, enabling pretrained LLMs to understand and generate geological evolution trends using natural language [50]. This strategy effectively addresses the limitations of traditional models in information expression and interpretability.

Overall, unlike traditional prediction methods that are directly trained on structured data, the LLM-based prediction paradigm first transforms multisource heterogeneous information into a unified natural-language representation and then employs tailored prompts to guide semantic reasoning and generative prediction. Moreover, after domain-knowledge embedding, LLMs can interpret specialized terminology and scientific concepts in IDRF with much higher accuracy, substantially enhancing semantic understanding and long-horizon reasoning—capabilities that are particularly critical for tasks such as aftershock-sequence forecasting, rainfall–runoff prediction, disaster-chain evolution inference, and infrastructure-response simulation. However, applying LLMs to predictive tasks still faces several practical challenges. Operationally, LLMs rely on high-quality annotated data and precise natural-language descriptions. For multimodal applications, several thousand cross-modal alignment samples are additionally needed. Data scarcity remains a fundamental bottleneck in the IDRF. From an application perspective, although LLMs are capable of generating seemingly coherent reasoning chains, their generative outputs lack explicit and interpretable probabilistic structures and do not incorporate physics-based constraints. IDRF scenarios often involve high-dimensional inputs, and directly embedding such data into an LLM may significantly increase the risk of hallucinations.

These limitations make it difficult for LLMs to meet the stringent reliability and verification requirements of disaster prediction tasks. A promising direction is to adopt a hybrid LLM-DL paradigm, in which the LLM performs data preprocessing, semantic reasoning, and interpretable analysis, while DL models carry out physically consistent prediction tasks. Alternatively, DL models can be used for dimensionality reduction and feature extraction, and only the semantic representations are fed into the LLM to avoid the curse of dimensionality. We emphasize that technological advancement is incremental, and rather than dismissing LLMs in prediction scenarios, it is more constructive to leverage their strengths.

This paper proposes a generalized technical framework designed to enable LLMs to perform prediction-oriented tasks in IDRF (see Fig. 6). This framework consists of the following components: (1) Prediction-task modeling: define prediction objectives, specify the input–output structure, and formulate the task in a manner interpretable to LLM based on domain requirements and contextual constraints; (2) Data collection: integrating structured and unstructured data sources and converting them into a unified representation suitable for LLM processing; (3) Language reconstruction: encoding training data into domain-specific natural-language descriptions to construct data-text pairs; (4) Model training: fine-tune a high-performance FM through supervised learning and reasoning-enhancement strategies to activate its predictive capabilities for domain-specific tasks; (5) Prompt-driven causal reasoning: develop task-specific prompts that guide the LLM to conduct explainable causal reasoning throughout the prediction process; (6) Prediction generation and output: execute prediction tasks using the optimized LLM to generate outputs such as trend forecasts, probabilistic assessments, or textual recommendations.

3.1.3. *LLMs empowering 'assessment' tasks*

Assessment tasks rely heavily on real-time post-disaster information, with the primary goal of accurately characterizing the damage state of infrastructure systems. Typical tasks include rapid damage identification, fault localization, functional-degradation assessment, and evaluation of disaster resistance [51–54]. However, disaster information is inherently uncertain, highly dynamic, and multisource heterogeneous, whereas infrastructure systems exhibit multi-mode and multi-scale patterns of complex failure behavior. These characteristics pose substantial challenges to assessment-oriented research in IDRF. Against this backdrop, LLMs—with their strong semantic understanding, cross-modal consistency modeling, and interpretable reasoning capabilities—offer a new technical pathway for integrating, abstracting, and assessing complex disaster information.

Integrating MLLMs with computer-vision techniques for real-time post-disaster field assessment represents an important frontier in current research. Wang et al. conducted a systematic evaluation comparing five MLLMs and five conventional DL models for crack detection in steel

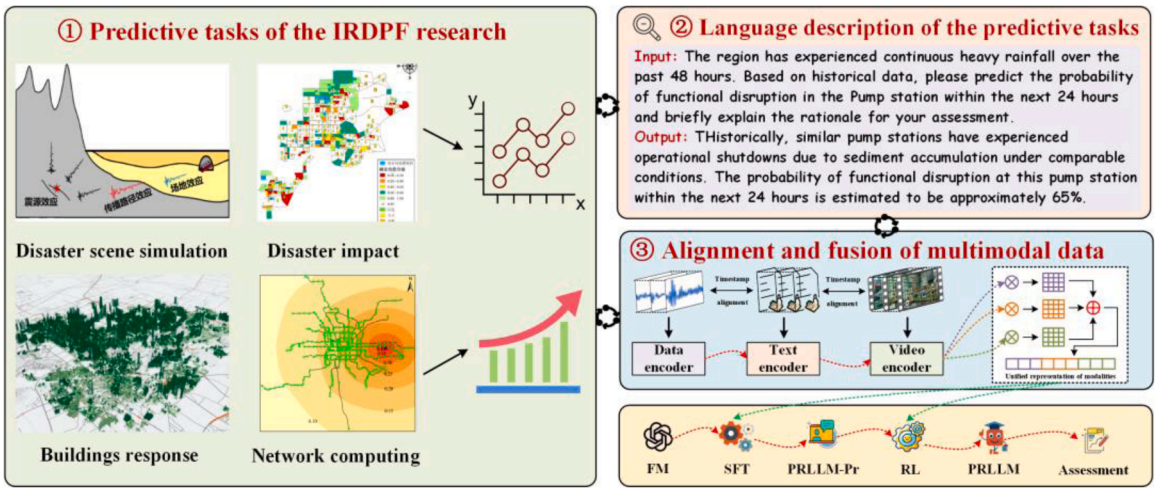

Fig. 6. Technical implementation framework for LLM-enabled predictive tasks,.

bridges. The results further validating the potential of LLMs in vision-based assessment tasks [55]. However, this approach relies primarily on the intrinsic semantic-understanding capabilities of high-performance LLMs without fine-tuning the model for structural damage scenarios; consequently, it still exhibits significant limitations in terms of reliability and engineering applicability. Zhang et al. noted that conventional DL models often lack the intelligence and reasoning capabilities required for real-world applications. To address this issue, they constructed a semantic dataset on wheat diseases using prompt engineering techniques and fine-tuned a vision-language model. The resulting domain-adapted model accurately classified wheat diseases and generated potential treatment suggestions, demonstrating strong applicability in real-world field conditions [56]. This provides a valuable reference for assessing infrastructure damage states using real-time ubiquitous monitoring data in post-disaster scenarios. An emerging research frontier involves applying LLMs to fault diagnosis and reliability analysis.LLMs can identify latent semantic associations and cross-modal patterns among fault-related features from heterogeneous data sources, thereby complementing traditional single-modal assessment paradigms. For instance, Lin et al. noted that conventional DL–based fault diagnosis methods are often tailored to specific scenarios and models, resulting in severe feature overlap and elevated misdiagnosis rates. To mitigate this issue, they proposed aligning descriptive engineering texts with equipment operational data through modality-aligned training. By integrating fuzzy semantic embeddings and learnable prompt embeddings, they fine-tuned a MLLM, resulting in significantly improved fault classification accuracy during experimental validation [57]. Pang et al. developed a dual-stage LLM architecture that integrates a self-supervised reconstruction network with a weakly supervised classification network. Systematic experiments on two real-world rotating machinery datasets demonstrated that the model achieved excellent diagnostic performance even under conditions of incomplete information, offering a promising technical pathway for reliability assessment [58].

In summary, LLMs exhibit promising potential in assessment-oriented tasks. By facilitating multimodal information integration and cross-scenario knowledge transfer, LLMs can support more comprehensive assessments in complex and highly uncertain post-disaster environments, particularly when combined with domain knowledge and task-specific models. For instance, treating DL models as external tools callable by LLMs and orchestrating them through an AI agent enable the combination of 'LLM-based reasoning and planning' with 'DL-based specialized recognition,' thereby enhancing system intelligence while maintaining engineering-level accuracy. However, consistent with the discussion in Section 3.1.2, LLM training depends heavily on high-quality, domain-specific annotated data—such as manually labeled damage grades, seismic-damage survey records, and annotated remote-sensing imagery. In the absence of industry-wide data-integration mechanisms, researchers often need to develop automated data-augmentation and expansion strategies (see Section 3.1.5) to alleviate performance bottlenecks caused by data scarcity.

This paper proposes a general technical framework for the rapid on-site assessment of infrastructure damage (see Fig. 7). The framework first constructs a multimodal semantic-alignment dataset by pairing infrastructure-damage images with their corresponding textual descriptions. Subsequently, a task-specific LLM is trained for damage identification through contrastive learning between visual and textual modalities, complemented by SFT, prompt engineering, and other techniques. Finally, by integrating a visualization interface and deploying the system on mobile devices, the framework enables efficient and intelligent on-site evaluation of infrastructure-damage conditions.

Beyond damage recognition, this technical framework can be extended to additional representative application scenarios, including: (1) Real-time disaster damage assessment from social-media streams: by enabling LLMs to rapidly extract, filter, and semantically interpret crowd-sourced posts, this approach supports quick identification of damage signals and emergent disruptions, allowing timely assessment of disaster impact under highly noisy social-media environments; and (2) Construction of Agentic AI systems for infrastructure resilience assessment: centered on a general-purpose LLM with domain-specific reasoning capabilities and built upon frameworks such as LangGraph, the system integrates multiple functional sub-models (e.g., data sensing, damage detection, functional analysis, and recovery simulation). It enables task autonomy, knowledge-driven reasoning, and collaborative interaction to support automated, streamlined, and intelligent resilience assessment for infrastructure systems.

3.1.4. LLMs empowering intelligent design, modeling and simulation

The design, modeling, and disaster simulation of infrastructure have long served as essential tools for reproducing complex physical phenomena, elucidating disaster mechanisms, assessing system vulnerabilities, and optimizing emergency response strategies. However, traditional modeling workflows are heavily dependent on manual operations—such as parameter configuration, geometric modeling, mesh generation, and plan drafting—which are time-consuming and labor-intensive. These processes also demand substantial domain expertise, design, and modeling experience, limiting their scalability for rapid response and high-throughput simulation tasks.

As a representative technology of Generative AI, LLMs exhibit significant potential to enhance model design, simulation, and system

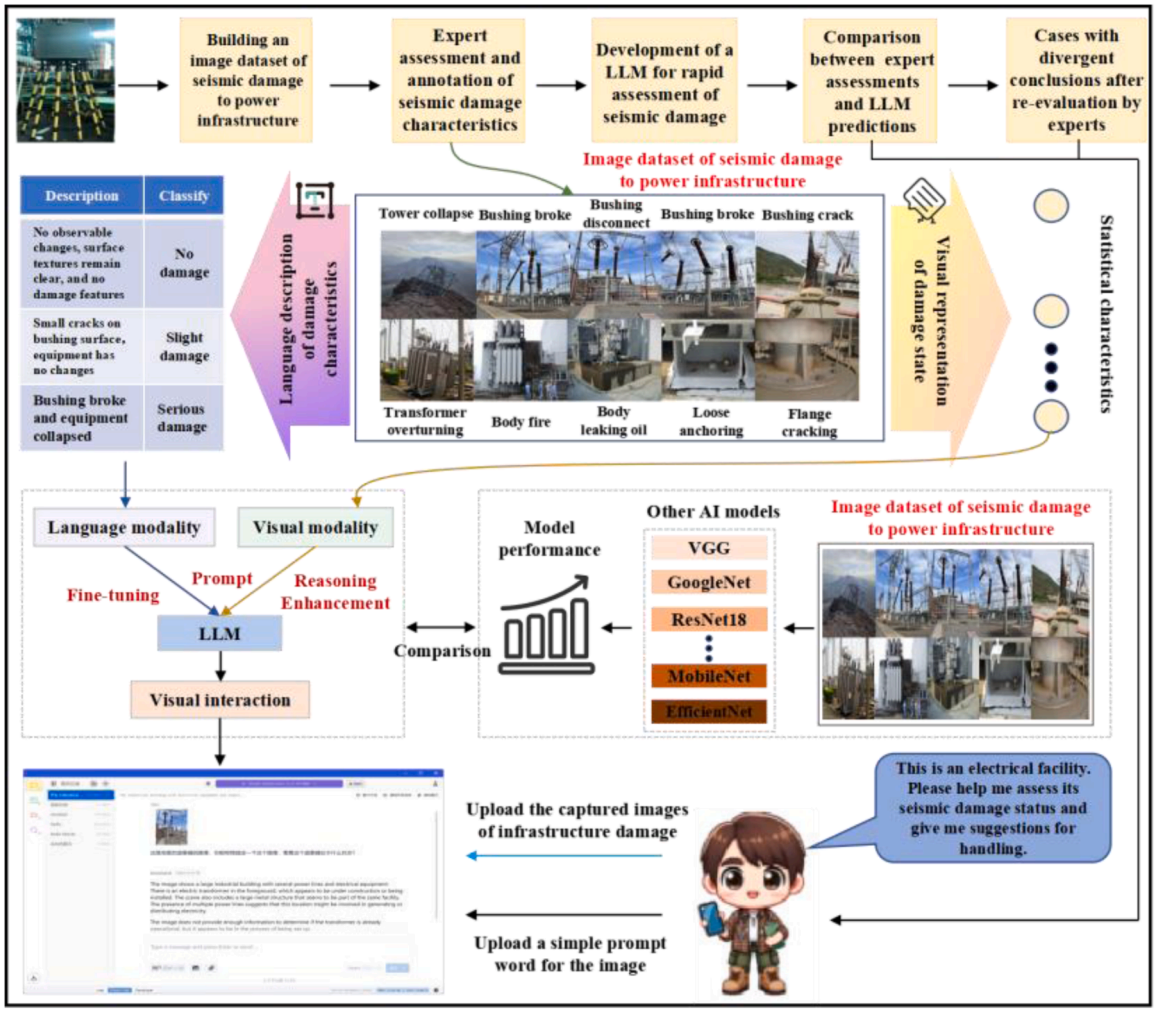

Fig. 7. Rapid on-site infrastructure damage assessment assisted by vision and natural language.

modeling tasks. The core concept involves enabling users to describe modeling requirements, design requirements, and control parameters using natural language. Based on semantic parsing and structural mapping rules, LLMs can then automatically generate modeling scripts, parameter configuration files, directly executable simulation code, and design scheme. These outputs can be interfaced with finite element analysis platforms, disaster simulation systems, or graph modeling engines, forming an intelligent closed-loop workflow from natural language input to model generation and simulation execution. Resilience-oriented system design is inherently a complex, multi-stage process. LLM-agent systems can integrate these otherwise fragmented components into a unified framework, enabling coherent, end-to-end collaborative design and reasoning. Dong et al. fine-tuned the Qwen2.5–7B model using 28,761 OpenFOAM configuration descriptions in natural language, enabling the automatic generation of executable CFD configurations from plain language prompts [59]. This result indicates that, with appropriate fine-tuning on high-quality domain-specific corpora, lightweight LLMs can also acquire the capability to assist in complex modeling and numerical simulation. Jiang and Zhang developed an LLM-based system for automated building-energy modeling. This platform can generate energy models with diverse geometric configurations, thermal-zoning layouts, material properties, and operational strategies directly from natural-language descriptions [60–64]. Their findings demonstrate that LLM-driven automated modeling and simulation substantially reduce manual workload, enhance efficiency and scalability, and enable non-expert users to participate in model construction through simple natural-language instructions. Furthermore, Liang et al. integrated large vision–language models with topology-optimization methods, enabling the optimization process—under LLM-guided gradient directions—to simultaneously incorporate human design preferences and structural-performance objectives. Their results show that this emerging intelligent-design paradigm fundamentally transforms the traditional workflow of 'CAD-based geometric modeling → CAE-based performance simulation → iterative design refinement,' paving the way toward a new generation of collaborative and generative structural-design methodologies [65].

In summary, LLM-based intelligent design, modeling, and simulation exhibit clear advantages in terms of low entry barriers, high efficiency, and strong generality. These models can also be integrated into DT systems, infrastructure resilience assessment platforms, and intelligent simulation frameworks, thereby promoting an end-to-end workflow that transitions from language-driven modeling to automated intelligent simulation. Nevertheless, several limitations remain. (1) LLMs are fundamentally language models and thus struggle to directly process highly structured engineering inputs such as geometric parameters, boundary conditions, and loading configurations, often requiring manual preprocessing or external toolchains. (2) LLMs cannot independently execute complex simulations or optimization procedures; instead, they must rely on APIs of established commercial software, most of which are closed-source, limiting deep coupling between the model and simulation engines. (3) At the application level, there is currently no unified standard or regulatory framework for auditing the compliance

and code-conformity of LLM-generated engineering designs and models. Physics-consistency checking, baseline comparison, and expert-in-the-loop review are essential measures for mitigating the risks associated with subtle or hidden errors. (4) Infrastructure systems are inherently complex, and theories and methodologies for resilience-oriented design remain under development. As a result, LLMs—without sufficient domain-knowledge constraints or physics-based guidance—face challenges in supporting high-fidelity engineering applications.

Despite these limitations, LLM-enabled design, modeling, and simulation hold substantial potential for IDRF. This paper proposes a reference technical framework to achieve LLM-driven intelligent infrastructure modeling and simulation. The workflow consists of the following steps: (1) selecting an open-source simulation platform and collecting a large set of simulation code files; (2) converting these files into professional natural-language descriptions to construct code–text pairs; (3) fine-tuning a domain-enhanced LLM to activate its capability for engineering model generation; (4) orchestrating the fine-tuned LLM with the simulation software's API through an agent framework and prompt engineering to build a standardized intelligent workflow; and enabling the LLM-driven agent system to perform end-to-end automated simulation, from natural-language task descriptions to simulation result outputs.

3.1.5. LLMs empowering data augmentation and enrichment

High-quality, multi-dimensional, and structured data form the essential foundation for model training, disaster reasoning, and intelligent decision-making. However, real-world engineering applications face persistent challenges, including data scarcity, high annotation costs, limited scenario diversity, and inadequate alignment across multimodal data. These limitations severely constrain the practical effectiveness and scalability of AI in tasks such as disaster identification, risk prediction, and post-disaster recovery modeling.

LLMs are fundamentally grounded in Generative AI. For example, CLIP can generate semantically aligned images from natural-language descriptions [66], whereas the Sora model can produce high-quality text-to-video content [67]. These advances suggest that LLM-based semantic augmentation can serve as a supportive tool for alleviating data scarcity, particularly by enhancing cross-modal consistency and annotation richness, rather than as a standalone solution. Yang et al. proposed a large model–based data augmentation framework in which semantic features are first extracted using CLIP and then combined with diffusion models to generate high-quality target images, thereby enabling a language-driven data generation pipeline [68]. Hsieh et al. emphasized that X-ray images, when detached from clinical context, offer limited diagnostic value. To address this limitation, they developed a multimodal modeling system that integrates structured patient data with contextual knowledge from radiology reports, leveraging LLMs to enhance data consistency and clinical reliability [69]. Collectively, these studies indicate that LLMs can facilitate semantic-driven data enrichment and cross-modal alignment, thereby complementing downstream model training with more structured and context-aware datasets.

Despite these advantages, the application of LLMs for data augmentation and enrichment faces several critical challenges. First, current LLMs are not originally designed for task-oriented data augmentation, and even trillion-parameter models struggle to reliably generate high-fidelity samples in highly specialized engineering domains. In many cases, lightweight and domain-specific models may provide a more practical solution. Second, hallucination remains an inherent risk of LLMs, as they are fundamentally next-token predictors. This risk cannot be eliminated through MLLMs alone and necessitates external grounding and verification mechanisms, such as physics-based constraints, deterministic simulations, and engineering-grade validation frameworks, to prevent dataset contamination. Third, while LLMs excel at natural-language generation, they remain limited in producing accurate numerical or structured engineering data. Hybrid generation pipelines that integrate LLMs with DL models specialized in feature extraction and structured-data synthesis may therefore improve overall reliability.

Nevertheless, we argue that LLM-enabled data augmentation and enrichment remain a crucial approach for addressing data scarcity in the era of AI. In the IDRF, typical cases include: (1) generating semantically consistent but linguistically diverse disaster reports for training models on event classification, sentiment analysis, and language comprehension; (2) simulating infrastructure damage states, degradation processes, and recovery trajectories under different hazard scenarios and generating structured textual samples for model training; (3) constructing traceable reasoning chains for causal inference enhancement; (4) transforming images, sensor outputs, or structural models into descriptive texts to build cross-modal datasets; (5) generating synthetic damage imagery, remote sensing data, or simulated sensor outputs via prompt-based text inputs for multimodal data expansion; and (6) leveraging the natural-language understanding capabilities of high-performance LLMs, the raw corpus—such as academic literature and textbooks—can be converted into instruction datasets suitable for SFT.

This paper proposes a physics-constrained MLLM–driven framework for augmenting infrastructure damage datasets, as illustrated in Fig. 8. The framework utilizes three core data types—real infrastructure images, corresponding finite element model images, and semantic description texts. Importantly, the FE simulations provide deterministic and physically grounded guidance on damage characteristics—such as crack location, orientation, and extent—ensuring reliability compared with purely probabilistic data-driven generation. A pretrained foundation model is then fine-tuned and enhanced with reasoning capabilities to enable data generation. In practice, users input an FE simulation image along with a well-designed textual prompt, and the MLLM generates a realistic damage image aligned with the input semantics. In this framework, the MLLM focuses on multimodal alignment and appearance synthesis, while the underlying damage morphology is constrained by deterministic FE modeling, thereby combining physical reliability with scalable data augmentation.

3.1.6. LLM-enabled embodied AI in the IDRF

Embodied AI represents a cutting-edge research frontier at the convergence of AI and robotics. Its core philosophy centers on the deep integration of perception, cognition, and action through dynamic physical interaction between intelligent agents and their environments, thereby enabling autonomous learning and continuous adaptation. As illustrated in Fig. 9, embodied AI is undergoing a paradigm shift—from perception-driven mechanisms to semantics-driven embodied large-model agents—fueled by the continuous integration of LLMs, robotic systems, and multimodal sensing platforms. In this framework, LLMs handle high-level task comprehension, semantic reasoning, and strategy formulation, while robotic systems engage in direct physical interaction with the environment, thereby establishing a closed-loop system of perception, cognition, and action. Embodied AI driven by LLMs is widely regarded as a crucial technological pathway toward achieving artificial general intelligence.

The development of efficient, intelligent, and multifunctional emergency response equipment has long been a key frontier in the IDRF. In recent years, technologies such as earthquake search-and-rescue robots, UAV-based sensing systems, and automated material delivery platforms have seen growing deployment in real-world disaster scenarios. Among these, unmanned systems serve as core components of action modules within future AI agent frameworks and are pivotal for realizing fully automated emergency response architectures. However, traditional unmanned systems typically rely on manual operation or rule-based programming, lacking true autonomous planning, intelligent decision-making, and adaptability to complex environments. The emergence of embodied intelligence offers a promising solution to these limitations. According to Liu et al., embodied AI integrates LLMs, world modeling mechanisms, and AI algorithms with robotic hardware, mechatronic systems, human–machine interfaces, and perception

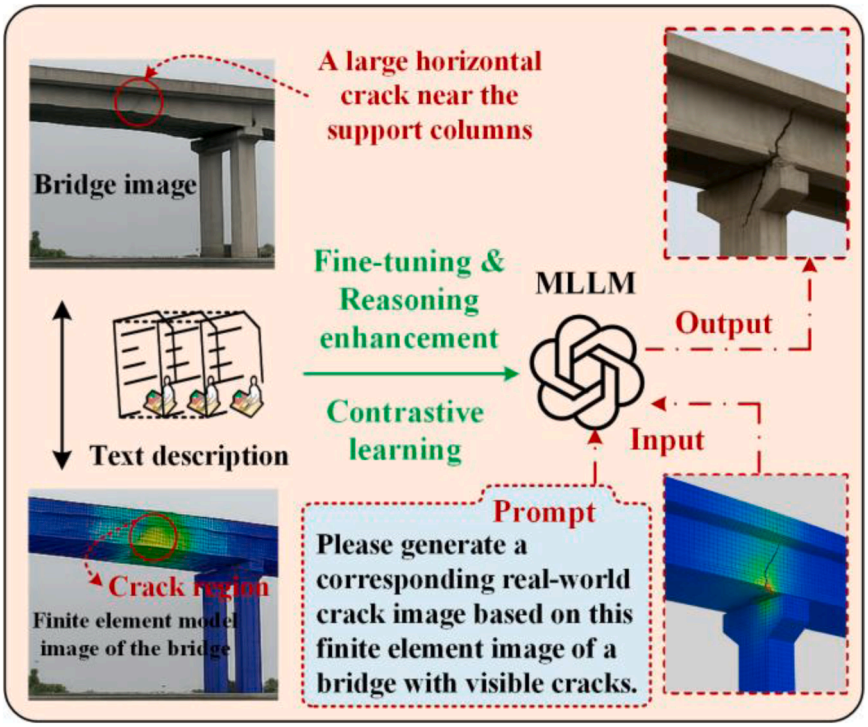

Fig. 8. Augmenting bridge damage image datasets using a MLLM.

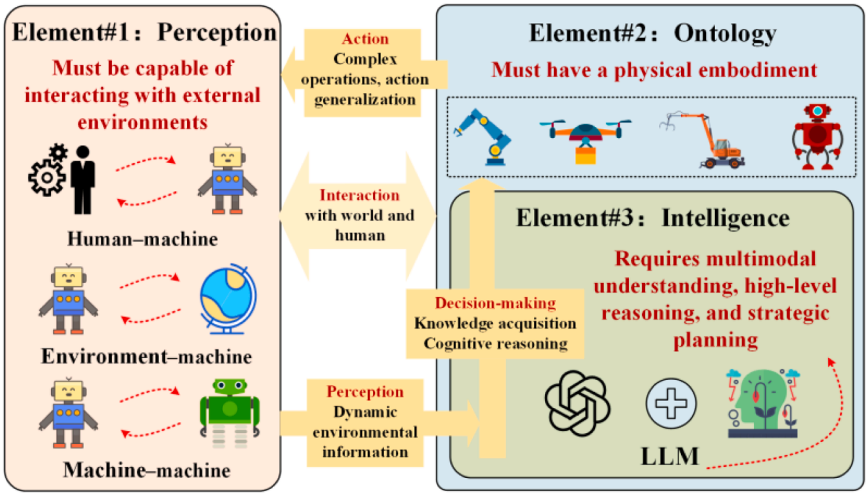

Fig. 9. LLM-powered embodied AI agent systems.

platforms—effectively endowing AI agents with physical embodiment and enabling deep interaction with the real world. This integration is propelling unmanned systems toward higher levels of autonomy and intelligence [70]. Tian et al. pointed out that by integrating capabilities such as perception, memory, reasoning, and tool use, LLMs can endow unmanned aerial vehicles with the essential characteristics of intelligent agents [71]. This finding demonstrates that LLM-driven UAVs can achieve higher-level situational understanding, task decomposition, and adaptive decision-making in disaster environments, thereby offering substantial potential for improving emergency inspection, disaster assessment, and autonomous navigation in damaged areas. In addition, the SaiV team conducted a series of 'ChatGPT-for-Robotics' experiments demonstrating that unmanned aerial vehicles can autonomously inspect wind-turbine blade cracks under natural-language instructions.. This study departs from traditional human-operated inspection workflows and highlights the remarkable performance and transformative potential enabled by integrating LLMs with robotic systems [72].

It is evident that integrating advanced natural-language understanding and action-execution capabilities into robotic systems has already demonstrated considerable potential within the IDRF. By further integrating the LLMs and AI agents developed in the preceding sections—equipped with prediction, assessment, design, simulation, and decision-making capabilities—with robotic platforms, intelligent unmanned systems, teleoperation frameworks, and multimodal-perception terminals, it becomes possible to develop embodied AI agents capable of natural-language interaction, environmental perception, and autonomous task execution. These agents can effectively support complex emergency response and disaster relief operations. For example, (1) inspection-oriented LLMs with semantic understanding capabilities can be integrated into robotic systems equipped with visual sensors, LiDAR,

infrared cameras, and other multimodal sensing technologies. Such agents can autonomously plan inspection routes and perform dynamic surveys of critical infrastructure—such as bridges, tunnels, and water treatment facilities—generating real-time reports on structural conditions; and (2) in complex disaster environments (e.g., earthquake debris fields or flood zones), embodied intelligent systems equipped with integrated perception, language understanding, and reasoning capabilities can operate UAVs or UGVs to autonomously transport emergency supplies, navigate hazardous zones, and clear obstructed access routes—thereby enhancing the efficiency of resource allocation and ensuring operational safety.

Despite these advancements, this line of application still faces numerous challenges. Although enabling robots to operate through natural-language instructions appears promising, the mapping from language to executable actions remains intrinsically challenging. Disaster scenarios often involve single-point or multi-point outbreaks, where on-site data streams are highly heterogeneous, multimodal, and uncertain, making real-time information interpretation a critical yet unresolved challenge. Moreover, emergency response demands low-latency and high-frequency reactions, whereas LLM inference inherently introduces delays, limiting their ability to support real-time control. Additional challenges—such as multi-device coordination, hardware-resource constraints, and limited cross-task generalization—further hinder the deployment of embodied AI in IDRF settings.

Nevertheless, we believe that technological progress is driven by continuous exploration and iteration. With ongoing advances in AI models, sensing technologies, edge computing, and robotic platforms, these challenges are expected to be gradually mitigated. The future of embodied intelligence in disaster-resilient infrastructure remains highly promising.

3.1.7. LLMs and AI agents empowering 'decision-support' tasks

Post-disaster emergency decision-making, intelligent urban operations, and resilience management of city-scale systems are representative 'decision-support' tasks within the IDRF. Traditional emergency response methods rely heavily on field investigations and manual judgment, which suffer from significant limitations in information acquisition, processing efficiency, and response speed [73,74]. The widely adopted paradigm of data-driven approaches, which integrate DL and optimization algorithms, essentially functions as an information processor and fails to fundamentally transform the cognitive logic and mechanisms of decision-making. In contrast, LLM-AI agents enable an end-to-end intelligent restructuring of the entire decision-making pipeline—from information comprehension and dynamic planning to autonomous decision-making and human-agent collaboration, thus offering a novel and highly effective paradigm for decision support in disaster scenarios.

Recent studies have provided strong evidence supporting the feasibility and advantages of this technological pathway. Shang et al. developed an intelligent-agent system based on an electronic health-record dataset, modeling clinical diagnosis as a multi-round interactive loop and dynamically generating treatment strategies through iterative querying and information integration [75]. Their findings demonstrate that, when supported by domain-specific datasets, LLMs can produce relatively reliable strategies through multi-round reasoning and debate, thereby providing effective decision support for healthcare professionals. Gao et al. developed the TxAgent system, which consists of three key components: a tool library containing 211 biomedical tools, a fine-tuned LLM for multi-step reasoning and tool execution, and an adaptive tool-retrieval module. By integrating reasoning, knowledge retrieval, and tool execution, the system is able to generate tailored treatment recommendations [76]. This work demonstrates how Agentic AI can effectively orchestrate a diverse set of tools, leveraging the LLM's domain expertise, reasoning capabilities, and workflow coordination to accomplish complex tasks through step-by-step processing and solution generation. These cases highlight the unique value of LLM-driven AI agents in decision-support applications.

Emergency response strategy is one of the most representative decision-support tasks within IDRF, involving a complex workflow that spans real-time disaster-information sensing, infrastructure-damage assessment, system-level functional-loss estimation, recovery-strategy optimization, resource-allocation planning, and solution generation. Agentic AI offers an integrated framework capable of merging these otherwise fragmented steps into a coherent decision-making pipeline, leveraging LLMs' capabilities in natural-language understanding, instruction following, and tool-chain execution to enable data interaction and human-machine collaboration. Fig. 10 illustrates a reference architecture and workflow for an Agentic AI–based post-disaster emergency decision-support system. At its core is a domain-adapted LLM equipped with multimodal perception, complex-task reasoning, and scenario-based planning and decision-making capabilities. Building on this foundation, the system incorporates specialized DL submodels, functional agents (as outlined in Sections 3.1.1–3.1.6), and unmanned platforms such as drones, autonomous vehicles, and teleoperation systems. Together, these components form an intelligent-agent system endowed with planning, perception, memory, action, and tool-invocation capabilities. Through the coordinated operation of modular components, the system realizes a closed-loop emergency-response process encompassing perception, recognition, planning, execution, memory, and reflection. This architecture enables rapid response and adaptive decision-making during sudden disasters, significantly enhancing the intelligence level of infrastructure emergency management.

In addition, the concept of developing a "Resilient City Smart Brain" has gained substantial traction within China's IDRF community. The overarching idea is to integrate multisource data—including seismic monitoring, infrastructure-performance sensing, disaster-assessment records, and historical hazard events—together with cross-sector data from meteorology, transportation, and urban-construction departments. On this basis, domain-specific large models, RAG-based knowledge bases, and specialized DL submodels can be trained for different application scenarios. Through multimodel coordination enabled by agent frameworks such as MCP, an intelligent-agent system for city-scale resilience management can be established. Furthermore, by incorporating DT technologies, this framework can evolve into a holistic 'Resilient City Smart Brain' capable of real-time monitoring, risk assessment, simulation-based prediction, emergency decision support, and optimized resource allocation. Such a system enables end-to-end intelligent collaboration across the entire disaster-prevention, mitigation, and resilience-enhancement workflow.

3.2. Case study on the application of LLMs and AI agents in IDRF

3.2.1. BDAChat: building damage rapid assessment based on visual language model

We present a representative case to demonstrate the advanced capabilities of LLMs in disaster assessment tasks. As illustrated in Fig. 11, we develops an object-centric rapid assessment framework that integrates high-precision building extraction, spatial–temporal pairing, and multi-task visual-language model (VLM) reasoning [77]. First, a SAM-based segmentation module, enhanced through pyramid decoding and multi-LoRA adaptation, is used to extract building footprints and post-disaster road networks with high geometric fidelity. Second, each building instance in pre- and post-disaster imagery is spatially paired using an expanded bounding-box strategy, allowing the model to capture contextual cues such as debris, flooding patterns, scorch marks, or structural disappearance. Third, a Video-LLaVA–style VLM is fine-tuned to perform three key assessment tasks: building-damage classification, disaster-type identification, and causal explanation generation. The model jointly reasons over spatial appearance changes, scene semantics, and multi-temporal contrasts, enabling interpretable and task-aligned assessments at city scale.

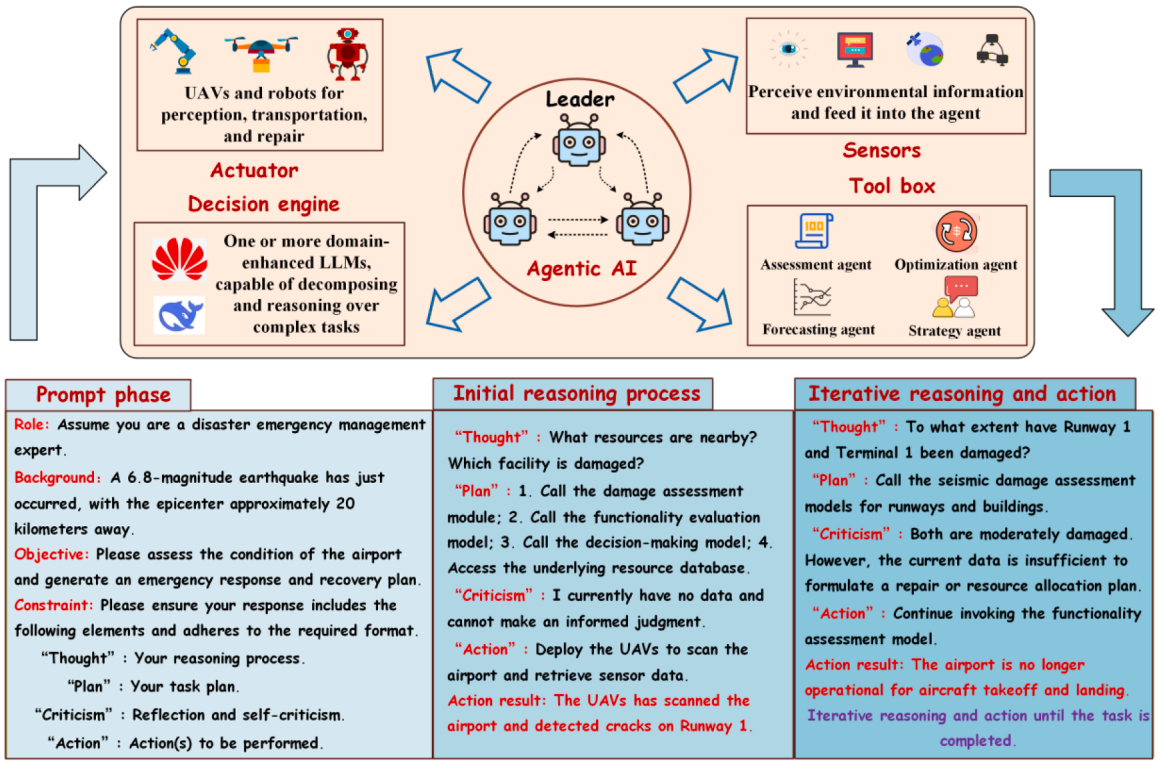

Fig. 10. LLM-AI-agent decision support system for infrastructure emergency management and its operational workflow.

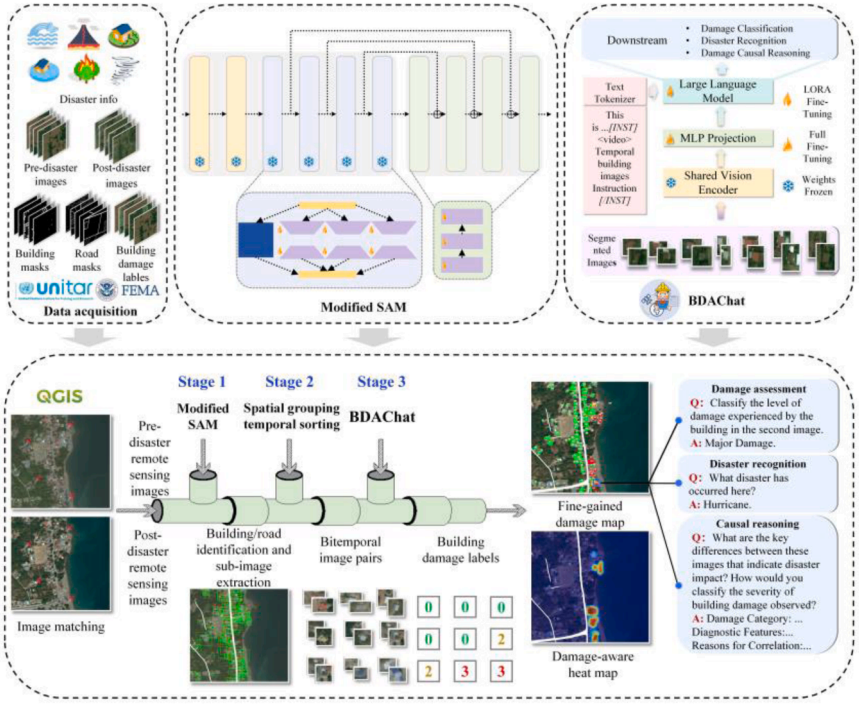

Fig. 11. The framework of the proposed three-stage pipeline for building damage assessment.

To verify applicability in real disaster contexts, the framework was deployed for the 2023 Lahaina wildfire event. A total of 1381 buildings were extracted automatically using the enhanced SAM module, producing clean object-level masks even under shadows, dense vegetation, and irregular urban layouts. Pre- and post-disaster tiles were generated at 1024×1024 resolution and paired for instance-level analysis, as shown in Fig. 12. When processed by the VLM, the system completed city-wide damage assessment within minutes, producing fine-grained four-level labels (No / Minor / Major / Destroyed). The output was consistent with official damage maps, while offering finer distinctions and richer causal diagnostics. For example, in cases where the official map labeled structures as 'undamaged,' the VLM correctly identified complete structural disappearance and material replacement as evidence of 'Destroyed,' providing clear natural-language reasoning for

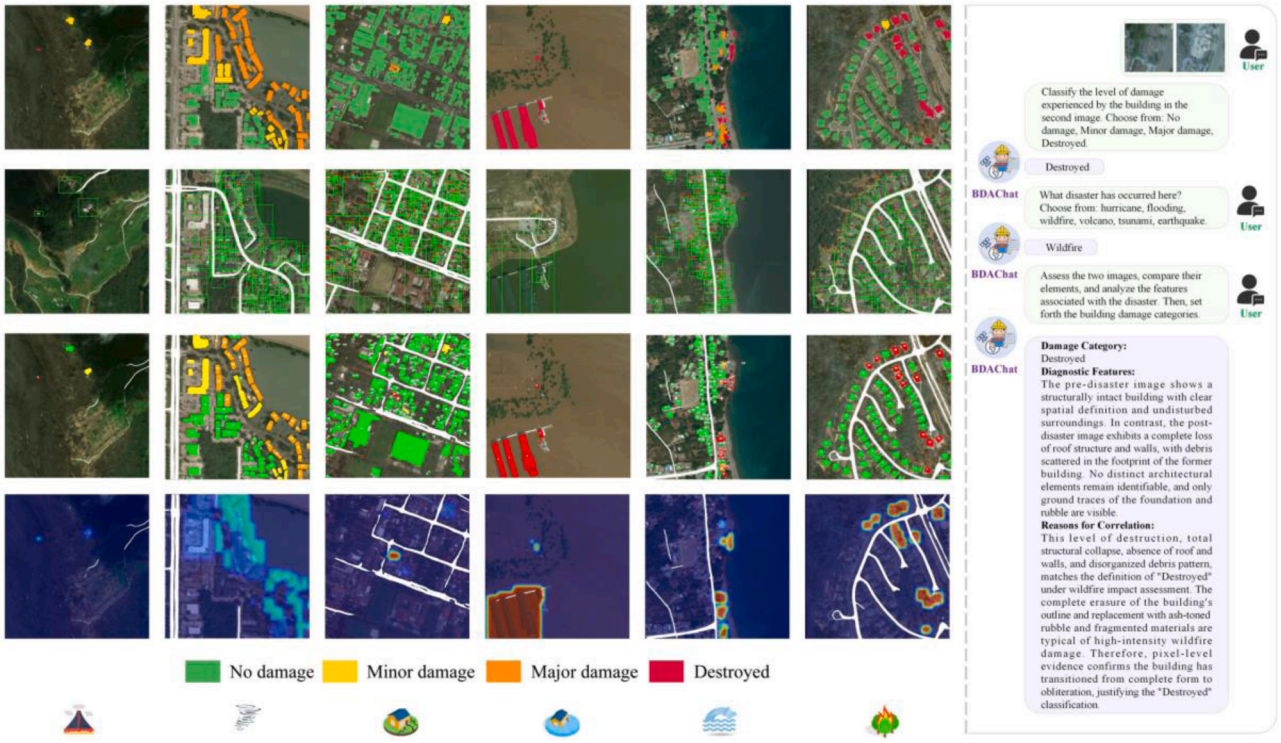

Fig. 12. Results of building damage assessment across disaster scenarios and Interpretive examples from BDAChat.

verification. The system further generated building-level damage maps and regional heat maps supporting emergency routing, resource allocation, and recovery planning.

The case study demonstrates that LLM/VLM-enabled assessment represents a significant advance over conventional computer-vision pipelines. First, the integration of natural-language reasoning allows the model to generate transparent, interpretable justifications—an essential requirement for engineering and governmental decision making. Second, VLMs provide strong cross-scenario generalization, enabling rapid adaptation to different hazards without retraining extensive task-specific models. Third, object-level reasoning over paired multi-temporal imagery produces more accurate and trustworthy assessments compared with single-frame or purely pixel-based methods.

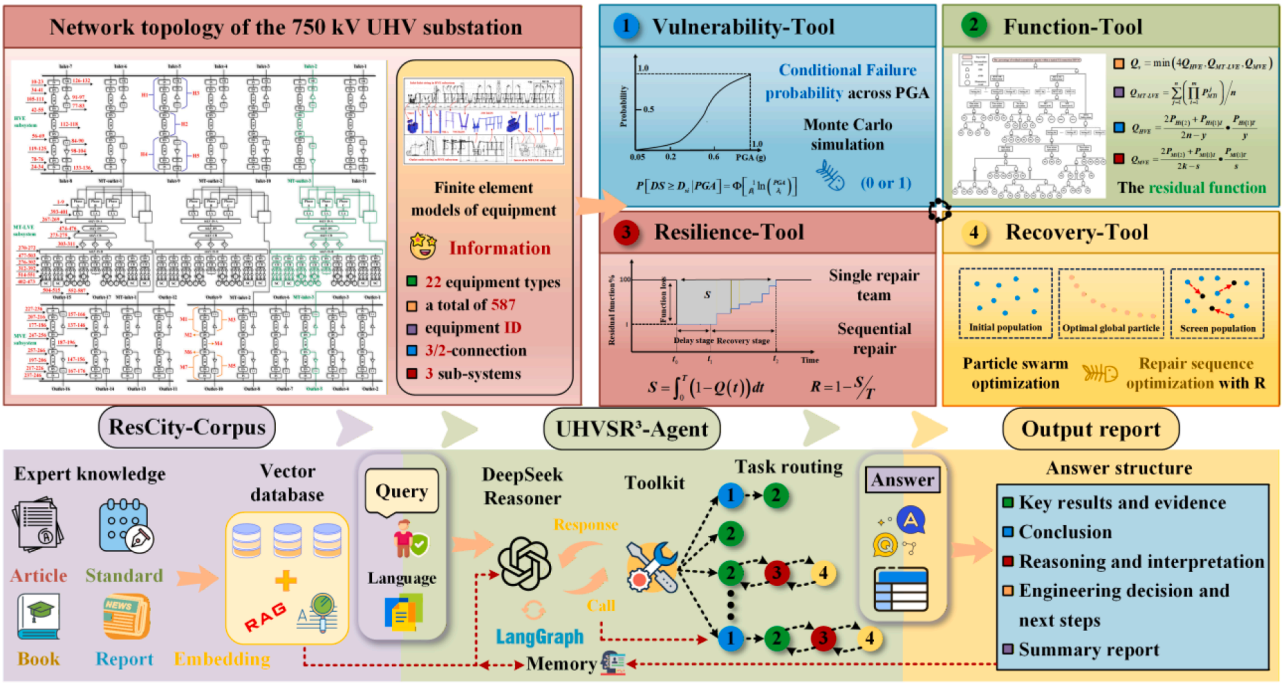

Fig. 13. The framework of the proposed UHVSR³-Agent for seismic resilience assessment.

3.2.2. UHVSR³-Agent: a LangGraph-based automated seismic resilience assessment agent for 750 kV ultra-high-voltage substations

To further demonstrate the unique value of AI agents in automated resilience assessment and decision support, an additional representative case study is presented. Based on LangGraph, we develop a physics-constrained reasoning agent framework, termed UHVSR³-Agent, with a language-controlled execution flow, for seismic resilience assessment and recovery planning of a 750 kV ultra-high-voltage (UHV) substation, as illustrated in Fig. 13. UHVSR³-Agent integrates multiple engineering-oriented tool modules. The vulnerability-Tool combines equipment-type–specific fragility curves with Monte Carlo simulation to generate post-earthquake device states. The function-Tool is based on a fault-tree representation of the substation, taking binary device states (0 for failed, 1 for intact) as inputs and producing the system-level functional indicator Q_s. The resilience-Tool evaluates the time-dependent functional recovery trajectory $Q(t)$ and the corresponding resilience index $R = 1 - \int_0^T (1 - Q(t))dt/T$, given an initial damage state and a specified repair sequence. The recovery-Tool formulates recovery planning as an optimization problem and employs a Particle Swarm Optimization to identify the repair sequence that maximizes the resilience index. In addition, a memory mechanism stores historical scenario configurations and validated results, enabling multi-round interactions with consistent physical assumptions. At the core of the agent lies the high-performance reasoning model DeepSeek-Reasoner, which is responsible for workflow orchestration and natural language generation. The overall process is organized as a directed state graph. Specifically, the LLM first interprets user queries and determines the routing and invocation order of engineering tools, then collects the corresponding computational results. At the reporting stage, an external expert knowledge base is queried to refine technical terminology and contextual explanations. All interpretations and recommendations are generated under strict evidence constraints, ensuring that the final outputs are structurally transparent, auditable, and physically consistent for resilience-informed decision support.

UHVSR³-Agent is capable of effectively addressing a wide range of questions related to functional assessment, seismic scenario simulation, seismic resilience evaluation, system functionality recovery processes, and optimal repair sequence optimization. To validate the practical applicability of this agent, a deterministic set of damaged components is used as input to query UHVSR³-Agent for the optimal repair strategy. As

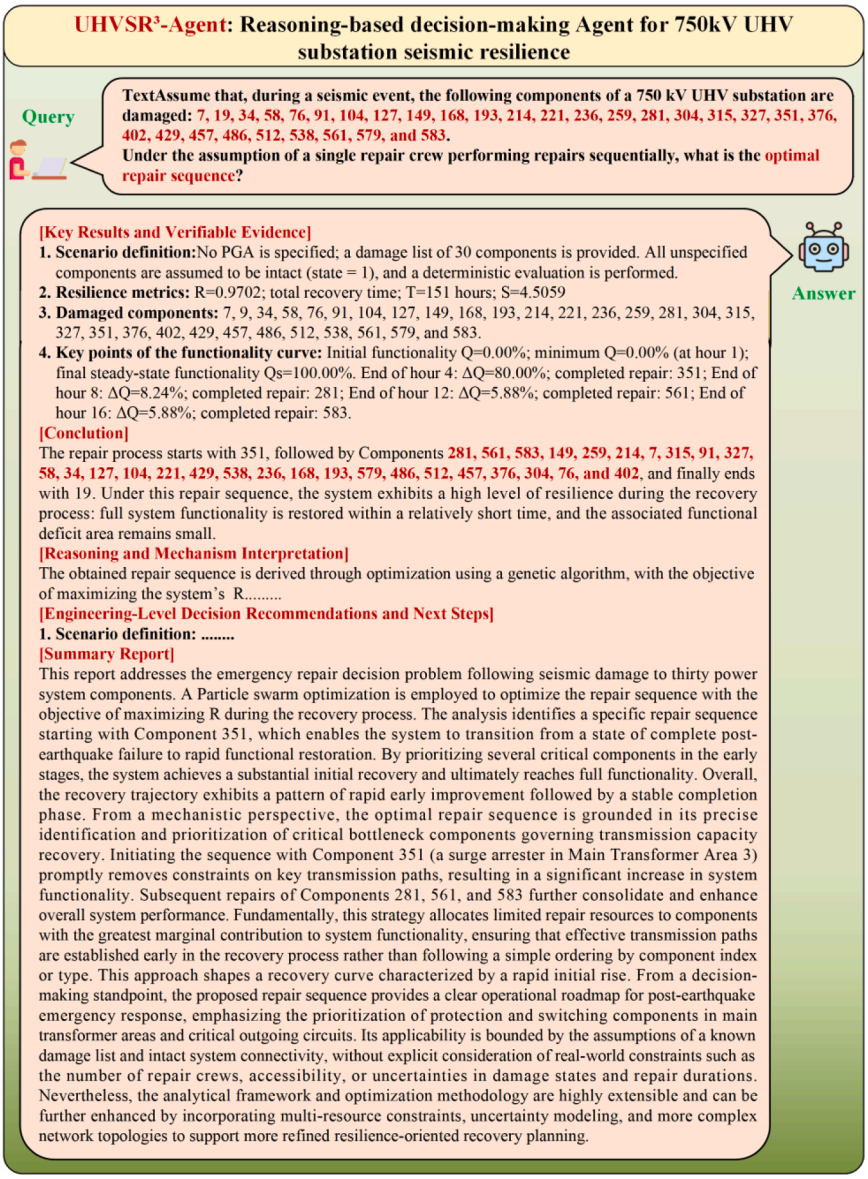

Fig. 14. Answer example of the seismic resilience assessment and recovery decision from UHVSR³-Agent.

illustrated in Fig. 14, UHVSR^3-Agent is able to perform effective semantic routing based on the query content, thereby invoking the appropriate physics-based and optimization tools to complete the task. UHVSR^3-Agent provides a structured response across five complementary dimensions: Key Results and Verifiable Evidence, Conclusion, Reasoning and Mechanism Interpretation, Engineering-level Decision Recommendations and Next Steps, and Summary Report.By using natural language as the entry point of the control flow, UHVSR^3-Agent not only delivers clear conclusions, but also provides a complete chain of verifiable evidence together with engineering-level decision support, enabling interpretable and auditable seismic resilience decision analysis.

This case study demonstrates the practical value of AI agents in infrastructure disaster resilience by showing how physics-based models, optimization tools, and reasoning modules can be coherently integrated through a language-controlled execution framework. By embedding deterministic engineering tools within an agentic workflow, UHVSR^3-Agent ensures physical consistency and engineering-level accuracy while enabling flexible task routing and adaptive analysis across diverse seismic resilience queries. Moreover, the use of natural language as the control interface, together with evidence-constrained reporting and memory-supported interactions, enhances interpretability, traceability, and usability for resilience-informed decision support. Overall, this case illustrates that physics-constrained AI agents can serve as an effective coordination and decision-support layer for automated, explainable, and actionable resilience assessment in IDRF.

4. Challenges and limitations

4.1. Data scarcity and security

Data scarcity represents one of the most fundamental and persistent challenges in applying LLMs and AI agents to IDRF tasks. Unlike other AI application domains that benefit from abundant, continuously updated, user-generated data, the IDRF is characterized by low-frequency, high-consequence disaster events, limited labeled datasets, and fragmented multi-agency data silos. These characteristics severely constrain the development, fine-tuning, and validation of domain-specific LLMs. Moreover, disaster-related datasets often lack comprehensive coverage across hazards, infrastructure types, and intensity levels. Many extreme or rare failure modes are underrepresented or entirely absent, resulting in substantial distribution gaps. The inherent heterogeneity of IDRF data—including remote-sensing imagery, sensor time-series, simulation outputs, and textual reports—further complicates the construction of standardized, large-scale, and high-quality training corpora.

To address data-related challenges, several feasible strategies can be pursued. These strategies include leveraging LLM- and physics-based approaches for data augmentation (Section 3.1.5), utilizing AI-agent driven corpus construction and automated annotation (Section 3.1.1), curating small but high-quality and authoritative multimodal benchmark datasets, and advancing techniques that fuse heterogeneous modalities to improve data efficiency. Furthermore, government-led institutional support is essential for enabling cross-agency data sharing and ensuring data security. This need is particularly critical in AI era. For example, China established the National Data Administration in 2023, tasked with coordinating the development of foundational data-governance frameworks, integrating and sharing cross-sector data resources, and promoting their effective utilization. Since its establishment, the agency has released several high-quality datasets across various industries, providing valuable support for AI model training and validation.

4.2. Ethical concerns, privacy, and fairness

Ethical and fairness concerns pose substantial challenges for deploying LLMs and AI agents in IDRF scenarios. IDRF'tasks influence public safety, emergency resource allocation, and the protection of vulnerable communities, making any model bias or hallucinated output potentially consequential. Due to data scarcity, uneven data distribution, and fragmented cross-agency information, LLMs may inadvertently propagate historical biases—such as unequal representation of regions or infrastructure types—resulting in skewed risk assessments or unfair decision recommendations. Additionally, the non-deterministic nature of LLM outputs and their lack of causal grounding complicate accountability in safety-critical workflows. Privacy risks also arise because disaster-related datasets often contain sensitive information on critical facilities, operational states, and population movements, which, if mishandled, may lead to security breaches or ethical concerns.

Several strategies can help mitigate these ethical and fairness-related challenges. These strategies include integrating fairness-assessment metrics into model development, adopting bias-mitigation techniques, and implementing human-in-the-loop oversight to ensure that LLM-generated insights remain advisory rather than authoritative. Privacy-preserving technologies—such as federated learning and on-premise deployment—can reduce the risk of sensitive data exposure. Furthermore, implementing auditable model logs, standardized accountability protocols, and cross-agency governance frameworks can enhance transparency and promote responsible AI use. Together, these measures help ensure that LLM technologies are deployed safely, fairly, and ethically in IDRF applications.

4.2. High training and deployment costs

Mainstream LLMs often contain tens or even hundreds of billions of parameters, rendering local deployment on personal computers virtually infeasible due to their heavy reliance on high-performance computing clusters. For example, deploying a full-scale open-source model such as DeepSeek-R1 locally requires at least 512 GB of RAM and eight NVIDIA A100 or H100 GPUs. Moreover, enhancing FMs through fine-tuning or inference-time augmentation requires high-quality corpus development, fine-grained annotation, and expert involvement—all of which incur significant upfront costs. In practical disaster scenarios, using LLMs and AI agents for online inference also requires high-performance hardware environments, including high-memory GPUs, robust bandwidth, and low-latency communication links. In post-disaster settings—such as remote mountainous areas, grassroots command centers, or edge devices—the lack of stable power and communication infrastructure poses major challenges for real-time deployment and operation of LLMs.

Addressing these challenges requires advances in model compression, resource allocation, and scalable system design rather than relying solely on large, general-purpose models. Promising directions include developing domain-specific lightweight models with fewer parameters but stronger alignment to IDRF's tasks; adopting parameter-efficient fine-tuning techniques to reduce computational overhead; and leveraging shared cloud-based or regional computing platforms to support training and inference. Furthermore, integrating edge computing with cloud–agent architectures can enable real-time disaster analysis even under constrained infrastructure conditions. At the policy level, establishing shared computing hubs, incentivizing open-source model development, and promoting cross-agency resource pooling can significantly lower deployment barriers.

4.4. Uncertainty, reliability, and operational risks

LLMs inherently produce probabilistic, non-deterministic outputs without explicit physical constraints, making predictive uncertainty difficult to quantify and output consistency challenging to ensure across different prompts, sampling strategies, or operating conditions. In high-stakes disaster scenarios, such uncertainties may propagate into hazard forecasting, damage assessment, and prioritization workflows, increasing the risk of misjudgment or delayed responses. Hallucinations—fluent yet incorrect or physically implausible outputs—pose

additional reliability risks, particularly when models encounter out-of-distribution events, incomplete observations, conflicting multimodal cues, or noisy field data. Beyond these model-level issues, LLMs and AI agents introduce new system-level failure modes, including inherent bias propagation, adversarial prompt manipulation, cascading errors in multi-agent systems, and mis-coordination caused by incorrect task decomposition or tool invocation. In multi-agent settings, a single hallucinated or biased output may be amplified downstream, triggering error cascades that undermine overall system safety. Collectively, these uncertainties and emerging risks represent significant challenges for the trustworthy deployment of LLM-driven IDRF systems.

Mitigating these risks requires reliability safeguards at both the model and system-engineering levels. Useful strategies include applying uncertainty-quantification techniques—such as ensemble prompting, robustness testing, and confidence calibration—to generate more reliable uncertainty indicators. Hybrid frameworks that couple LLM reasoning with physics-based simulations or DT models can improve output consistency and reduce hallucination risks. Human-in-the-loop mechanisms should be embedded to ensure that LLMs serve strictly as decision-support assistants. At the system level, operational safeguards—such as auditable agent logs, anomaly-detection modules, multilayer verification pipelines, and scenario-based evaluation protocols—are essential for identifying and correcting erroneous outputs. Finally, incorporating fault-tolerant mechanisms, including redundant reasoning paths, multi-model cross-validation, and agent consensus protocols, can further enhance the robustness and operational safety of LLM–agent systems in complex disaster environments.

5. Conclusions and future outlook

This paper introduces a unified cognitive–action framework that positions LLMs and AI agents as central enablers of next-generation disaster-resilient infrastructure. By mapping the capabilities of LLM-Agentic AI to the full spectrum of IDRF tasks—from knowledge anchoring and predictive reasoning to multimodal assessment, intelligent simulation, and decision support—we show how these technologies can fundamentally reshape disaster-prevention workflows. The case studies further highlight their potential for transparent, scalable, and high-impact real-world applications.

Looking ahead, future research should move beyond incremental extensions of existing methodologies and instead focus on advancing the next generation of intelligent resilience technologies. Several strategic directions warrant sustained long-term attention.

(1) Beyond physics-informed models: toward science-grounded cognitive AI. LLMs should evolve from correlation-driven reasoning toward hybrid neuro-symbolic and physics-grounded cognition, enabling explicit causal explanations, uncertainty-aware decisions, and verifiable robustness under extreme or previously unseen hazards.
(2) Domain-governed lightweight FMs for real deployment. Developing sub-14B LLMs is not only a technical optimization challenge but also a governance imperative—requiring new pipelines for curated data management, model distillation, and secure on-premise deployment across government agencies and infrastructure operators.
(3) Foundation-level multimodal benchmarks for resilience science. Rather than task-specific datasets, IDRF needs shared cross-hazard, cross-infrastructure, and cross-scale multimodal benchmarks that support standardized evaluation, scientific reproducibility, and community-wide methodological convergence.
(4) Convergence of embodied intelligence, digital twins, and agentic planning. Future systems should integrate LLM-driven cognitive modules with real-time world models, UAV/UGV robotics, and edge computing to form closed-loop embodied agents capable of autonomous inspection, intervention, and adaptive response in highly dynamic disaster environments.
(5) City-scale Agentic-AI ecosystems and resilient governance architectures. A 'Resilient City Smart Brain' requires not only advanced AI capabilities but also institutional coordination, standardized data infrastructures, interoperable DT platforms, and multi-agent governance mechanisms to support whole-of-city intelligence.

Overall, LLM-Agentic AI represents more than an incremental improvement; it opens a pathway toward autonomous, explainable, and self-evolving disaster-resilience technologies. Realizing this vision will require coordinated advances in AI methodologies, infrastructure engineering, data governance, and interdisciplinary resilience science.

CRediT authorship contribution statement

Jiawei Cui: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Changhai Zhai:** Supervision, Funding acquisition. **Yong Wang:** Visualization, Supervision. **Yuhao Li:** Validation.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, "**Large language models and AI agents in disaster-resilient infrastructure: concepts, applications, pathways, and challenges.**"

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Data availability

No data was used for the research described in the article.

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